

Games in Normal Form

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Game Theory: Introduction

Game theory is the multi-agent generalization of (normative) decision theory.

In **non-cooperative games**, self-interested, rational agents take decisions, maximizing their expected utility, that also depend on the behavior of the others. In **cooperative (or coalition) games**, agents take decision as a whole (groups or *teams*) in order to maximize a common utility function.

Normal-form games (usually) represent *simultaneous games*, in which players take decisions without knowing those of the others.

The **extensive form** (game tree) is preferred to represent *sequential games* (players choose strategies in some order, i.e. additional information).

Normal-Form Games: definitions and representations

Definition

A finite N -person **normal-form game** (or *strategic-form game*) is a tuple $(\mathcal{N}, A, u)$ in which:

- $\mathcal{N}$ is a finite set of N players;
- $A = A_1 \times \dots \times A_N$, where A_i is a finite set of *actions* (or *pure strategies*) available to player i ;
- $u = (u_1, \dots, u_N)$ where $u_i : A \rightarrow \mathbb{R}$ is a real-valued *utility* (or *payoff*) function for player i .

The vector $a = (a_1, \dots, a_N) \in A$ is called *action profile*.

Solution concepts: mathematical formalization of reasoning leading to some specific outcomes in multi-agent decision problems.

Normal-Form Games: definitions and representations

In **two-player games** with finite action sets, the normal form is represented by a **payoff bi-matrix**:

	C_1	C_2	$\dots$	C_{m_C}
R_1	r_{11}, c_{11}	r_{12}, c_{12}	$\dots$	r_{1m_C}, c_{1m_C}
R_2	r_{21}, c_{21}	r_{22}, c_{22}	$\dots$	r_{2m_C}, c_{2m_C}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
R_{m_R}	r_{m_R1}, c_{m_R1}	r_{m_R2}, c_{m_R2}	$\dots$	$r_{m_Rm_C}, c_{m_Rm_C}$

where $u_1(a) = r_{11}$, and $u_2(a) = c_{11}$ are the payoffs (utility value) associated with the action profile $a = (R_1, C_1)$.

Classification of two-player two-action games

	C_1	C_2
R_1	a, A	c, C
R_2	b, B	d, D

Two-player two-action normal-form games

Definition

In a two-player game of ***pure competition*** players have exactly opposed interests, i.e. $\exists c$ such that $\forall a \in A = A_1 \times A_2, u_1(a) + u_2(a) = c$ (if $c = 0$, called *zero-sum games*).

Example

MATCHING PENNIES

	Head	Tail
Head	1, -1	-1, 1
Tail	-1, 1	1, -1

ROCK-PAPER-SCISSORS

	Rock	Paper	Scissors
Rock	0, 0	-1, 1	1, -1
Paper	1, -1	0, 0	-1, 1
Scissors	-1, 1	1, -1	0, 0

Two-player two-action normal-form games

Definition

A game is **symmetric** if payoffs depend only on the actions played, not on who is playing them. (Simplified representation with $R_i = C_i = A_i$.)

	A_1	A_2
A_1	a, a	c, b
A_2	b, c	d, d

In **common-payoff** games, players have exactly the same interests, i.e. $\forall a \in A, \forall i, j \in \mathcal{N}, u_i(a) = u_j(a)$.

Example (TRAFFIC HANDEDNESS)

	Left	Right
Left	1, 1	0, 0
Right	0, 0	1, 1

Two-player two-action normal-form games

Definition

A 2×2 **coordination game** is a two-player two-strategy game in which it is beneficial to play in the same way. The payoff matrix satisfies $a > b, d > c$ and $A > C, D > B$ (if symmetric, $a > b \geq d > c$).

	C_1	C_2
R_1	a, A	c, C
R_2	b, B	d, D

	A_1	A_2
A_1	a, a	c, b
A_2	b, c	d, d

Example (STAG HUNT)

	Stag	Hare
Stag	3, 3	0, 2
Hare	2, 0	1, 1

Two-player two-action normal-form games

Definition

A 2×2 **anti-coordination game** is a two-player two-strategy game in which it is beneficial to play different strategies. The payoff matrix satisfies $b > a, c > d$ and $B > D, C > A$ (if symmetric, $b > a \geq c > d$).

	C_1	C_2
R_1	a, A	c, C
R_2	b, B	d, D

	A_1	A_2
A_1	a, a	c, b
A_2	b, c	d, d

Example (HAWK DOVE/CHICKEN)

	Dove	Hawk
Dove	3, 3	1, 4
Hawk	4, 1	0, 0

	Swerve	Straight
Swerve	0, 0	-1, 1
Straight	1, -1	-100, -100

Two-player two-action normal-form games

Some games mix features of coordination and competition.

Example (BATTLE OF SEXES)

Two friends are deciding to attend a movie together. Row player likes Action movie more than Comedy, Column player has the opposite preference. Both of them prefer to spend time together instead of watching their favorite movie alone.

	Action	Comedy
Action	2, 1	0, 0
Comedy	0, 0	1, 2

Coordination: because they would prefer to go to the same movie.

Competition: because they have conflicting preferences.

Two-player two-action normal-form games

Example (PRISONER'S DILEMMA)

Two thieves have been captured and are questioned separately in different rooms. They can either stay silent (Cooperate) or betray the other (Defect). If both criminals betray the other (Defect, Defect), each of them serves three years in prison. If player 1 betrays player 2 but player 2 stays silent, player 1 will be set free and player 2 will serve four years in prison (and vice versa). Finally, if both players remain silent, both of them will serve only one year in prison because of a minor charge.

	Cooperate	Defect	$\implies$		C	D
Cooperate	-1, -1	-4, 0		C	3, 3	0, 4
Defect	0, -4	-3, -3		D	4, 0	1, 1

Symmetric PD games satisfy $b > a > d > c$.

Pareto optimality

Definition

Given a normal-form game $(\mathcal{N}, A, u)$, the action profile a **Pareto dominates** action profile a' if $\forall i \in \mathcal{N}, u_i(a) \geq u_i(a')$ and $\exists j \in \mathcal{N}$ such that $u_j(a) > u_j(a')$. A strategy profile a is **Pareto optimal**, or strictly Pareto efficient, if $\nexists a' \in A$ that Pareto dominates a .

Remark

Every game has at least one Pareto optimal outcome.

Example

- In the Battle-of-Sexes, the profiles (A,A) and (C,C) are Pareto optimal.
- In the Prisoner's Dilemma, all but (D,D) are Pareto optimal profiles.
- In zero-sum games, all profiles are Pareto optimal.

Strategic dominance

Definition

Let $a_i, a'_i \in A_i$ be two actions available to player $i \in \mathcal{N}$, then

- a_i *strictly dominates* a'_i if $\forall a_{-i} \in A_{-i}, u_i(a_i, a_{-i}) > u_i(a'_i, a_{-i})$;
- a_i *weakly dominates* a'_i if $\forall a_{-i} \in A_{-i}, u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i})$ and $\exists \tilde{a}_{-i} \in A_{-i}$ such that $u_i(a_i, \tilde{a}_{-i}) > u_i(a'_i, \tilde{a}_{-i})$;
- a_i *very weakly dominates* a'_i if $\forall a_{-i} \in A_{-i}, u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i})$.

A pure strategy is (strictly) **dominant** for an agent if it (strictly) dominates any other pure strategy for that agent.

A pure strategy $a_i \in A_i$ is (strictly) **dominated** for agent $i \in \mathcal{N}$ if $\exists a'_i \neq a_i \in A_i$ that (strictly) dominates a_i .

Remark

Rational players would never play dominated strategies. But dominance does not depend on the rationality of other players.

Iterated dominance

Example

Consider the two-player game with payoff matrix

	<i>L</i>	<i>C</i>	<i>R</i>
<i>U</i>	3, 1	2, 1	0, 0
<i>M</i>	1, 1	1, 1	5, 0
<i>D</i>	0, 1	4, 1	0, 0

Some strategies become dominated only after the elimination of other dominated ones.

	<i>L</i>	<i>C</i>	<i>R</i>
<i>U</i>	3, 1	2, 1	0, 0
<i>M</i>	1, 1	1, 1	5, 0
<i>D</i>	0, 1	4, 1	0, 0

$\Rightarrow$

	<i>L</i>	<i>C</i>
<i>U</i>	3, 1	2, 1
<i>M</i>	1, 1	1, 1
<i>D</i>	0, 1	4, 1

$\Rightarrow$

	<i>L</i>	<i>C</i>
<i>U</i>	3, 1	2, 1
<i>D</i>	0, 1	4, 1

Iterated dominance

Is the process of ***iterated elimination of strictly dominated (pure) strategies (IESDS)*** univocally defined?

Proposition (*Church-Rosser property*)

Every strategy that is strictly dominated at some stage of the process would still be dominated at any later stage if reintroduced.

The set A_i^∞ of pure strategies of agent i surviving the IESDS does not depend on the order of elimination.

A game can be solved using IESDS.

Definition

(a) A game is *solvable* by iterated strict dominance if for each player i , A_i^∞ is a singleton. (b) A strategy profile a is a ***strict dominance equilibrium*** if a_i is a strictly dominant strategy $\forall i \in \mathcal{N}$.

(b) implies (a) but (a) does not (always) imply (b).

Best-response

Suppose that agent i knows what the other players will play (i.e. a_{-i}), then i can reply choosing the action that maximizes her utility.

Definition

Player i 's **best response** to the action profile a_{-i} is a strategy $a_i^* \in A_i$ such that $u_i(a_i^*, a_{-i}) \geq u_i(a_i, a_{-i})$, $\forall a_i \in A_i$, or equivalently such that

$$a_i^* \in B_i(a_{-i}) = \arg \max_{a_i} u_i(a_i, a_{-i}).$$

Example

- In the Stag Hunt game, S (resp. H) is best response to S (resp. H)
- In the Hawk-Dove game, H (resp. D) is best response to D (resp. H);
- In the Prisoner's Dilemma, D is best response to both C and D;
- In Matching Pennies, H (resp. T) is best response to H (resp. T) for one player and viceversa for the other.

Exercise (COURNOT MODEL, 1838)

Two firms compete for the production of the same good. The (continuous) actions of the firms are the produced quantities q_1, q_2 , whereas the price of the good is set by the market-clearing condition, i.e. $P(Q) = \max \{0, a - Q\}$, where $Q = q_1 + q_2$ is the total quantity present on the market and $a > 0$ is the maximum produced quantity. The utility of firm i is given by

$$u_i(q_i, q_j) = q_i P(Q) - C(q_i) = \max \{0, a - Q\} - cq_i,$$

where $0 < c < a$ is the marginal production cost.

- 1 Show that IESDS identifies a possible solution of the game.
- 2 Show that the best-response relations identify the same solution.

Pure-strategy Nash equilibrium

Definition

An action profile $a^* = (a_1^*, a_2^*, \dots, a_N^*)$ is a **pure-strategy Nash equilibrium** for a (normal-form) game $G = (\mathcal{N}, A, u)$ if $\forall i \in \mathcal{N}$,

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a'_i, a_{-i}^*), \quad \forall a'_i \in A_i.$$

Remark

A pure N.E. is a fixed-point of best-response, i.e. $a_i^* \in B_i(a_{-i}^*)$, $\forall i \in \mathcal{N}$. Informally, a N.E. is a strategy profile from which no player finds profitable to *unilaterally deviate*.

Example

- In the Battle of Sexes, the pure-strategy N.E. are (A,A) and (C,C).
- In the Prisoner's Dilemma, the only pure-strategy N.E. is (D,D).
- Matching Pennies has no pure-strategy N.E.

Exercise (HOTELLING'S LOCATION MODEL)

Consumers are uniformly distributed along a unit interval (e.g. beach).
Two firms (e.g. icecream vendors) selling the same product are located at $x_1 = 0$ and $x_2 = 1$.

The marginal cost of producing the good is $c > 0$.

There is a transportation cost t per unit distance charged to the consumers. Consumers have unit demand.

The strategies of vendors are prices p_1, p_2 .

What is the unique pure-strategy Nash Equilibrium of the game?

Pure-strategy Nash equilibrium

When does a game admit at least one Nash equilibrium in pure strategies?

Proposition

The game $(\mathcal{N}, A, u)$ has a Nash equilibrium in pure strategies if the following conditions are satisfied $\forall i \in \mathcal{N}$,

- *the set A_i of actions available to player i is a non-empty, compact, convex subset of a Euclidean space;*
- *utility functions u_i are continuous in A and quasi-concave in A_i .*

The proof exploits fixed-point theorems for set-valued functions (e.g. conditions on actions and utility satisfy the hypothesis of the fixed-point theorem).

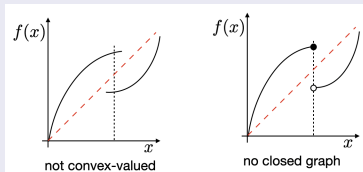
Pure-strategy Nash equilibrium

Theorem (KAKUTANI FIXED-POINT TH.)

Let X be a compact convex subset of $\mathbb{R}^N$ and let $f : X \rightarrow X$ be a set-valued function for which

- $\forall x \in X$, the set $f(x)$ is non-empty and convex-valued, i.e. $\forall x \in X$, $f(x)$ is a convex set;
- the graph of f is closed, i.e. for all sequences $\{x_n\}_{n \geq 0}$, $\{y_n\}_{n \geq 0}$ with $y_n \in f(x_n) \forall n$, such that $x_n \rightarrow x$ and $y_n \rightarrow y$ for $n \rightarrow \infty$, it also holds that $y \in f(x)$.

Then there exists $x^* \in X$ such that $x^* \in f(x^*)$.



IESDS vs. pure-strategy Nash equilibria

Proposition

In a finite game, if the IESDS leaves a unique action profile a^ , then a^* is the unique Nash equilibrium of the game.*

Proposition

If a^ is a Nash equilibrium, then actions $a_1^*, \dots, a_N^*$ have to survive the IESDS process.*

Assumptions have to be slightly reinforced to extend such properties to infinite games.

Proposition

Let $(\mathcal{N}, A, u)$ be a game in which A_i is compact and u_i is continuous for every $i \in \mathcal{N}$, then if a strict dominance solution of the game exists, it is also a Nash equilibrium.

Mixed strategies

Definition

A **mixed strategy** for player i is a probability measure $\sigma_i \in \Sigma_i = \mathcal{P}(A_i)$ s.t. $\sigma_i : a_i \mapsto \sigma_i(a_i)$ i.e. that maps action $a_i \in A_i$ on its probability $\sigma_i(a_i)$ to be played by player i .

The *support* of the strategy σ_i is the set $\{a_i | \sigma_i(a_i) > 0\}$.

Definition

The **mixed-strategy profile** $\sigma \in \Sigma$ with $\sigma = (\sigma_1, \dots, \sigma_N)$ and $\Sigma = \Sigma_1 \times \dots \times \Sigma_N$, is the *product measure* $\sigma = \prod_{i \in \mathcal{N}} \sigma_i$.

Definition

Given a normal-form game $(\mathcal{N}, A, u)$, the **expected utility** for a player i of a mixed strategy profile $\sigma = (\sigma_1, \dots, \sigma_N)$ is

$$U_i(\sigma) = \sum_{a \in A} u_i(a) \prod_{j \in \mathcal{N}} \sigma_j(a_j).$$

Strategic dominance in mixed strategies

Definition

Let $s_i, s'_i \in S_i$ be two (pure/mixed) strategies of player $i \in \mathcal{N}$, then

- s_i *strictly dominates* s'_i if $\forall s_{-i} \in S_{-i}, u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$;
- s_i *weakly dominates* s'_i if $\forall s_{-i} \in S_{-i}, u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ and $\exists \tilde{s}_{-i} \in S_{-i}$ such that $u_i(s_i, \tilde{s}_{-i}) > u_i(s'_i, \tilde{s}_{-i})$;
- s_i *very weakly dominates* s'_i if $\forall s_{-i} \in S_{-i}, u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$.

A strategy is (strictly, weakly, very weakly) **dominant** for an agent if it (strictly, ...) dominates any other strategy for that agent.

A strategy $s_i \in S_i$ is (strictly, ...) **dominated** for agent $i \in \mathcal{N}$ if $\exists s'_i \neq s_i \in S_i$ that (strictly, ...) dominates s_i .

Remark

Operatively complicated because based on comparing s_i, s'_i against all, possibly mixed, strategy profiles of the others.

Strategic dominance in mixed strategies

Proposition

A pure strategy $a_i \in A_i$ strictly dominates $a'_i \in A_i$ if and only if

$$\forall a_{-i} \in A_{-i}, \quad u_i(a_i, a_{-i}) > u_i(a'_i, a_{-i}).$$

Remark

A pure strategy a_i which is not dominated by any other pure strategy can still be dominated by a mixed strategy $\sigma_i \in \Sigma_i$. Again, this condition can be verified checking only against other players' pure strategies, i.e. verifying that

$$\forall a_{-i} \in A_{-i}, u_i(\sigma_i, a_{-i}) > u_i(a_i, a_{-i}).$$

Strategic dominance in mixed strategies

Example

Consider the game described by the following payoff table

	l	r
U	6, 0	0, 6
M	0, 6	6, 0
D	2, 0	2, 0

Strategy D is not dominated by pure strategies, but by e.g. $\sigma_R = (\frac{1}{2}, \frac{1}{2}, 0)$.
Indeed, $u_R(\sigma_R, l) = 3 > u_R(D, l) = 2$ and $u_R(\sigma_R, r) = 3 > u_R(D, r) = 2$.

Strategic dominance in mixed strategies

Remark

Any mixed strategy which puts positive probability on a dominated pure strategy is itself dominated. But a mixed strategy that puts weight only on undominated pure strategies can still be dominated.

Example

Consider the game with payoff table

	l	r
U	6, 0	0, 6
M	0, 6	6, 0
D	4, 0	4, 0

U and M are not dominated by D , but their mixture $\sigma_R = (\frac{1}{2}, \frac{1}{2}, 0)$ is dominated by D , because $u_R(\sigma_R, x) = 3 < 4 = u_R(D, x)$ with $x = l, r$. More generally, $\sigma_R = (p, 1 - p, 0)$ is dominated $\forall p \in (0, 1)$.

Strategic dominance in mixed strategies

Definition (IESDS process)

First set $A_i^0 = A_i$ and $\Sigma_i^0 = \Sigma_i$, then recursively define

$$A_i^t = \left\{ a_i \in A_i^{t-1} \mid \nexists \sigma_i \in \Sigma_i^{t-1} : \forall a_{-i} \in A_{-i}^{t-1}, \quad u_i(\sigma_i, a_{-i}) > u_i(a_i, a_{-i}) \right\}$$

and $\Sigma_i^t = \{ \sigma_i \in \Sigma_i : \text{supp}(\sigma_i) \subset A_i^t \}$. The set $A_i^\infty = \bigcap_{t=0}^\infty A_i^t$ is the set of pure strategies of agent i that survive the IESDS. Then, given

$$\hat{\Sigma}_i^\infty = \{ \sigma_i \in \Sigma_i : \text{supp}(\sigma_i) \subset A_i^\infty \},$$

in order to eliminate possibly dominated mixed strategies present in the last set, one finally defines

$$\Sigma_i^\infty = \left\{ \sigma_i \in \hat{\Sigma}_i^\infty \mid \nexists \sigma'_i \in \hat{\Sigma}_i^\infty : \forall a_{-i} \in A_{-i}^\infty, \quad u_i(\sigma'_i, a_{-i}) > u_i(\sigma_i, a_{-i}) \right\}.$$

Best Response and Nash Equilibria in mixed strategies

Definition

Player i 's **best response** to the strategy profile s_{-i} is a (possibly mixed) strategy $s_i^* \in S_i$ such that $u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i})$, $\forall s_i \in S_i$.

Definition

A **mixed-strategy Nash equilibrium** of a normal-form game $(\mathcal{N}, A, u)$ is a Nash equilibrium of its mixed strategies, i.e. a mixed-strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*)$ such that $\forall i \in \mathcal{N}$,

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i', \sigma_{-i}^*), \quad \forall \sigma_i' \in \Sigma_i.$$

Remark

The definition is operatively almost useless because there is an infinity of possible mixed strategies.

Best Response and Nash Equilibria in mixed strategies

A useful implication:

Proposition

Let $G = (\mathcal{N}, A, u)$ be a finite strategic (or normal-form) game. Then $\sigma^* \in \Sigma$ is a mixed-strategy Nash equilibrium of G if and only if $\forall i \in \mathcal{N}$, $\forall a_i \in \text{supp}(\sigma_i^*)$ is such that $a_i \in B_i(\sigma_{-i}^*)$.

Remark

It follows that every action in the support of any player's mixed-strategy equilibrium yields that player the same payoff.

For two-player games, we can identify mixed-strategy N.E. by using the method of ***equating opponent's payoffs***.

Best Response and Nash Equilibria in mixed strategies

Example

In the BoS game, consider strategy profile (σ_R, σ_C) with $\sigma_R = (p, 1 - p)$ and $\sigma_C = (q, 1 - q)$. The indifference condition on opponent's pure-strategy payoffs gives:

$$u_C(\sigma_R, A) = u_C(\sigma_R, C) \implies p = 2(1 - p) \implies p^* = \frac{2}{3}$$

$$u_R(A, \sigma_C) = u_R(C, \sigma_C) \implies 2q = 1 - q \implies q^* = \frac{1}{3}.$$

The profile $(\sigma_R^*, \sigma_C^*) = (p^*, q^*) = (\frac{2}{3}, \frac{1}{3})$ is a mixed-strategy N.E. It is Pareto dominated by both pure-strategy Nash equilibria,

$$u_R(\sigma_R^*, \sigma_C^*) = 2p^*q^* + (1 - p^*)(1 - q^*) = 2/3$$

$$u_C(\sigma_R^*, \sigma_C^*) = 1p^*q^* + 2(1 - p^*)(1 - q^*) = 2/3.$$

Existence of mixed-strategy Nash Equilibria

Theorem (NASH, 1950)

Every finite normal-form game has a mixed-strategy Nash equilibrium.

Proof. (Hint).

For a generalized game with $\tilde{A}_i = \Sigma_i$, all conditions of existence of a Nash equilibrium in pure strategies are met: (1) $\forall i \in \mathcal{N}$, the set of mixed strategies $\Sigma_i = \left\{ \sigma_i \mid \sum_{a \in A_i} \sigma_i(a) = 1 \text{ and } \sigma_i(a) \geq 0, \forall a \in A_i \right\}$ defined over a finite set A_i is a *non-empty, compact, convex* subset of a Euclidean space (it is a *simplex*); (2) the expected utility is linear in the probabilities, thus *quasi-concave* and *continuous* on them. $\square$

Proposition

If s^ is a Nash equilibrium, then the strategies $s_1^*, \dots, s_N^*$ have to survive the process of iterated elimination of strictly dominated strategies.*

Maxmin and Minmax strategies

Example

Consider the two-player game with payoff table

	L	R
U	2, 1	2, -20
M	3, 0	-10, 1
D	-100, 2	3, 3

(D, R) is a N.E. but it is risky, the profile (U, L) is safer for both.

Definition

In a normal-form game $G = (\mathcal{N}, A, u)$, the **maxmin strategy** s_i^* for player i is defined by

$$s_i^* \in \arg \max_{s_i} \min_{s_{-i}} u_i(s_i, s_{-i}),$$

$\underline{v}_i = \max_{s_i} \min_{s_{-i}} u_i(s_i, s_{-i})$ is the *maxmin value* (or *security level*) for i .

Maxmin and Minmax strategies

Definition

In a normal-form game $G = (\mathcal{N}, A, u)$, the **minmax strategy** for player i against players $-i$ is

$$s_i^* \in \arg \min_{s_i} \max_{s_{-i}} u_{-i}(s_i, s_{-i})$$

and the *minmax value* of players $-i$,

$$\bar{v}_{-i} = \min_{s_i} \max_{s_{-i}} u_{-i}(s_i, s_{-i}).$$

Definition

In n -player games, the minmax strategy for player i against player j is i 's component of $s_{-j}^* = \arg \min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$. The minmax value for player j is $\min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$.

Maxmin and Minmax strategies

Proposition

In any (two-player) normal-form game, a player's maxmin value is at most equal to her minmax value, i.e. $\underline{v}_i \leq \bar{v}_i$.

Definition

If any (two-player) normal-form game admits $\underline{v}_i = \bar{v}_i, \forall i \in \mathcal{N}$, then the payoff $\underline{v}_1 = \bar{v}_1 = v$ is called the *value of the game*. The corresponding maxmin and minmax strategies are called **optimal strategies**.

Proposition

*Given a **zero-sum two-player game**, then for $i \neq j$,*

$$\underline{v}_j = \max_{s_j} \min_{s_i} u_j(s_i, s_j) = - \min_{s_j} \max_{s_i} u_i(s_i, s_j) = -\bar{v}_i,$$

and $s_j^ \in \arg \max_{s_j} \min_{s_i} u_j(s_i, s_j)$ iff $s_j^* \in \arg \min_{s_j} \max_{s_i} u_i(s_i, s_j)$.*

Zero-sum games: Minimax Theorem

Theorem (v. NEUMANN, 1928)

For every finite two-player zero-sum game it exists a value v of the game in mixed strategies.

In terms of Nash equilibria, we can state the following equivalent result.

Theorem

Given a finite two-player zero-sum game,

- *if (s_i^*, s_j^*) is a N.E., then s_i^* is a maxmin strategy for player i and s_j^* is a maxmin strategy for player j ;*
- *if (s_i^*, s_j^*) is a N.E., then*
$$\max_{s_i} \min_{s_j} u_i(s_i, s_j) = \min_{s_j} \max_{s_i} u_i(s_i, s_j) = u_i(s_i^*, s_j^*)$$
and thus all N.E. yield the same payoffs (i.e. $\underline{v}_i = \bar{v}_i = v$);
- *if $\max_{s_i} \min_{s_j} u_i(s_i, s_j) = \min_{s_j} \max_{s_i} u_i(s_i, s_j)$ and s_i^* is a maxmin strategy for i and s_j^* is a maxmin strategy for j , then (s_i^*, s_j^*) is a N.E.*

Zero-sum games: N.E. as saddle points

Example

Consider the game with payoff table

	α	β	γ	δ	ϵ
A	0	-30	-20	-10	0
B	30	0	-10	0	10
C	20	10	0	10	20
D	10	0	-10	0	30
E	0	-10	-20	-30	0

pure-strategy N.E. are found solving $u_i(a_i^*, a_j^*) = \max_{a_i} \min_{a_j} u_i(a_i, a_j)$.
It gives $\max_{a_i} \{-30, -10, 0, -10, -30\} = 0$ and (C, γ) is the unique N.E.

Remark

Pure (resp. mixed) N.E. in zero-sum games are *saddles* of the payoff matrix (resp. utility function), and all saddles give the same payoff value, which is the value of the game.

Zero-sum games: N.E. as saddle points

Example

Consider the game with payoff table

	l	r
U	2	-1
D	1	2

Given strategies $\sigma_1 = (p, 1 - p)^T$ and $\sigma_2 = (q, 1 - q)^T$,

$$u_1(p, q) = \sigma_1^T W \sigma_2 = 4pq - q - 3p + 2$$

has a saddle point at $(p, q) = (1/4, 3/4)$, which identifies a mixed N.E.
Indeed, equating opponent's payoffs:

$$\begin{aligned} u_2(\sigma_1, l) = u_2(\sigma_1, r) &\implies -p - 1 = 3p - 2 \implies p = 1/4 \\ u_1(U, \sigma_2) = u_1(D, \sigma_2) &\implies 3q - 1 = 2 - q \implies q = 3/4 \end{aligned}$$